EE1060: Differential Equations and Transform Techniques - Quiz 1

Name: ______________________________

Roll No: ______________________________

Instructions

1. The quiz duration is 30 minutes, starting at 11:15. Ensure that you return the paper by 11:45. Please be aware that **late submissions** may incur **penalties**.
2. Go through all the questions carefully. Any misinterpretations will **not** be considered. Write your solutions in the designated space provided for each question. Answers written outside these areas will not be accepted.

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1. (2 points) Determine if the differential equation $\cos x \frac{d^2y}{dx^2} + \frac{dy}{dx} - xy = 0$ is linear and identify its order, providing supporting arguments for linearity or otherwise.

Solution:

The given equation is a second-order linear ODE. To show that the ODE is linear, it is sufficient to show that $F(x, \alpha y_1 + \beta y_2, \alpha y_1' + \beta y_1' + \alpha y_1'' + \beta y_2'') = \alpha F(x, y, y', y'') + \beta F(x, y, y', y'')$. Substituting the linear combinations of y_1, y_2, and their derivatives into F gives

$$\begin{aligned} F(x, \alpha y_1 + \beta y_2, \alpha y_1' + \beta y_2', \alpha y_1'' + \beta y_2'') &= \cos x(\alpha y_1'' + \beta y_2'') + (\alpha y_1' + \beta y_2') - x(\alpha y_1 + \beta y_2) \\ &= \alpha(\cos x\, y_1'' + y_1' - xy_1) + \beta(\cos x\, y_2'' + y_2' - xy_2) \\ &= \alpha F(x, y_1, y_1', y_1'') + \beta F(x, y_2, y_2', y_2'') \end{aligned}$$

Thus, the given differential equation is linear.

2. (2 points) Determine an explicit parametric solution for the ODE $\frac{dx}{dt} = -\frac{k}{x^2}$, where k is a constant.

Solution:

The given ODE can be written as $x^2\, dx = -k\, dt$, which is a separable ODE. Integrating on both sides gives

$$\int x^2\, dx = \int -k\, dt \quad \Rightarrow \quad \frac{x^3}{3} = -kt + C. \tag{1}$$

The explicit solution is thus

$$x(t) = (-3kt + 3C)^{1/3}. \tag{2}$$

3. (2 points) Determine the values of α and β for which $y = ce^{-bx} + \alpha x + \beta$ (where c is an arbitrary constant) is a solution to the ODE $\frac{dy}{dx} = ax - by$.

Solution:

Differentiating $y = ce^{-bx} + \alpha x + \beta$ with respect to x gives

$$\frac{dy}{dx} = -bc\, e^{-bx} + \alpha \tag{3}$$

Substituting (3) into the given ODE i.e., $\frac{dy}{dx} = ax - by$ and simplifying results in

$$-bc\, e^{-bx} + \alpha = ax - bc\, e^{-bx} - b\, \alpha x - b\beta \tag{4}$$

Comparing the terms, we find $\alpha = \frac{a}{b}$ and $\beta = 0$.

4. (2 points) Determine the maximum value of r for which $y = x^r e^{\alpha x}$ is a solution to the differential equation $\frac{d^2y}{dx^2} + 2\alpha \frac{dy}{dx} + \alpha^2 y = 0$.

Solution:

The first and second derivatives of $y = x^r e^{\alpha x}$ are

$$\frac{dy}{dx} = \left(\alpha x^r + r x^{r-1}\right) e^{\alpha x} \text{ and } \frac{d^2y}{dx^2} = \left[\alpha^2\, x^r + 2\alpha r\, x^{r-1} + r(r-1)\, x^{r-2}\right] e^{\alpha x} \tag{5}$$

Substitute y, y', and y'' into the differential equation and simplifying results in

$$\begin{aligned} \left[\left(\alpha^2\, x^r + 2\alpha r\, x^{r-1} + r(r-1)\, x^{r-2}\right) + \left(2\alpha^2\, x^r + 2\alpha r\, x^{r-1}\right) + \alpha^2 x^r\right] e^{\alpha x} &= 0 \\ \left(4\alpha^2\, x^2 + 4\alpha r\, x + r(r-1)\,\right) x^{r-2} e^{\alpha x} &= 0 \end{aligned} \tag{6}$$

Equation (6) holds only when $\alpha = 0$ and $r = 0$ or $r = 1$. Thus the maximum value of r is 1 (when $\alpha = 0$).

5. (2 points) Determine a one-parameter solution to the ODE $\frac{dx}{dt} + kx^2 \sin(bt) = 0$, where k and b are constants.

Solution:

The given ODE can be written as

$$\frac{dx}{x^2} = -k \sin(bt)\; dt \tag{7}$$

Integrating on both sides results in

$$-\frac{1}{x} = \frac{k}{b}\cos(bt) + C \tag{8}$$

Rearranging (8) gives

$$x(t) = -\frac{1}{\frac{k}{b}\cos(bt) + C} \tag{9}$$